

# Container-based High-Performance Computing Approaches for Simulating Large-Scale Biological Neural Networks

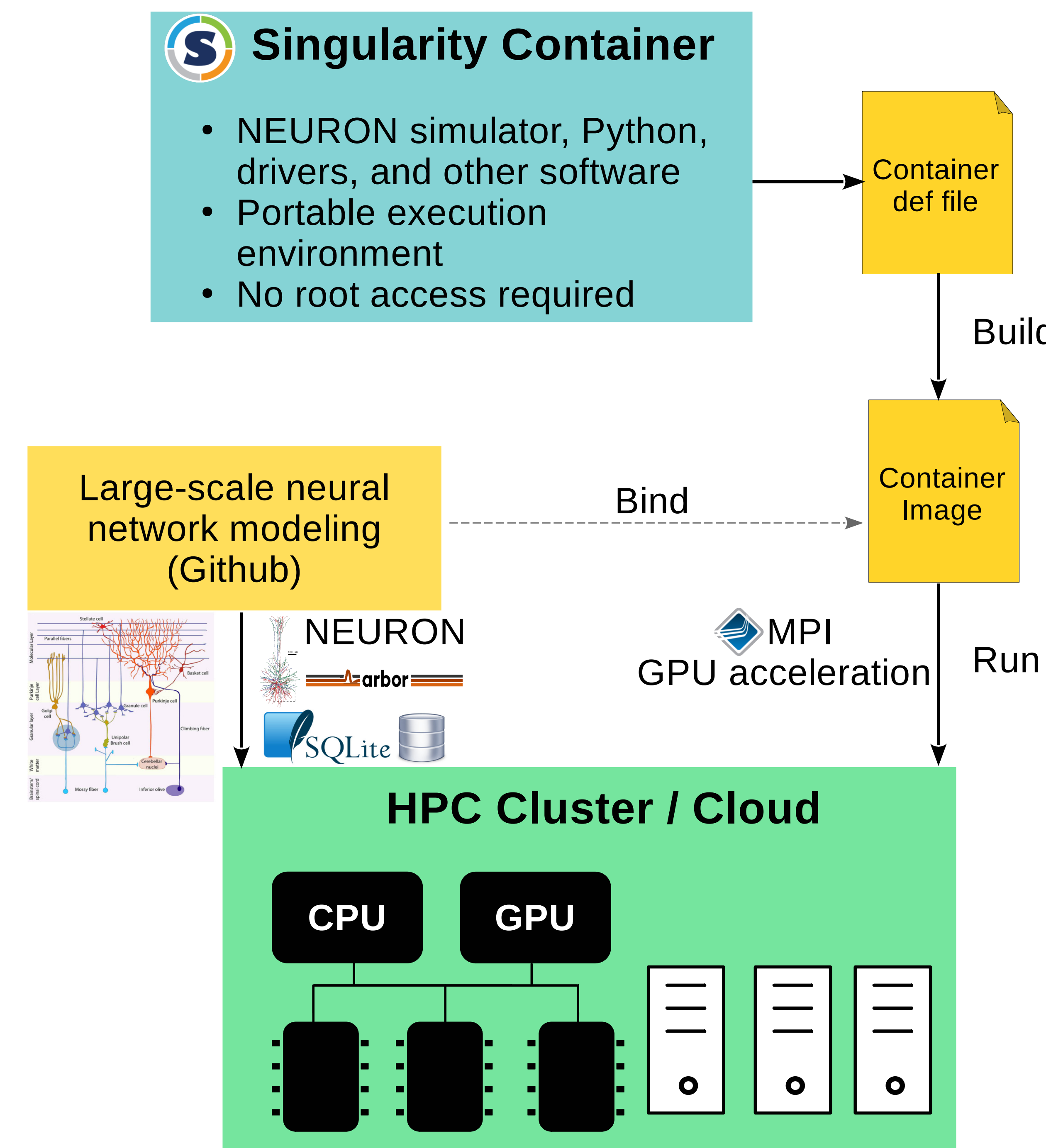
## Introduction

1. Modeling **large-scale, biophysically detailed neural networks** increasingly relies on **high-performance computing (HPC)**.
2. However, deploying simulations of such networks across diverse HPC environments remains challenging due to **software incompatibilities** and **complex dependencies**.
3. We address this issue using **Singularity containers** that encapsulate the entire simulation environment, enabling neuroscientists to achieve portability and reproducibility without requiring root privileges on HPC/Cloud systems.
4. We give practical demonstrations by benchmarking HPC environments with **cerebellar mossy fiber–granule cell network** simulations.

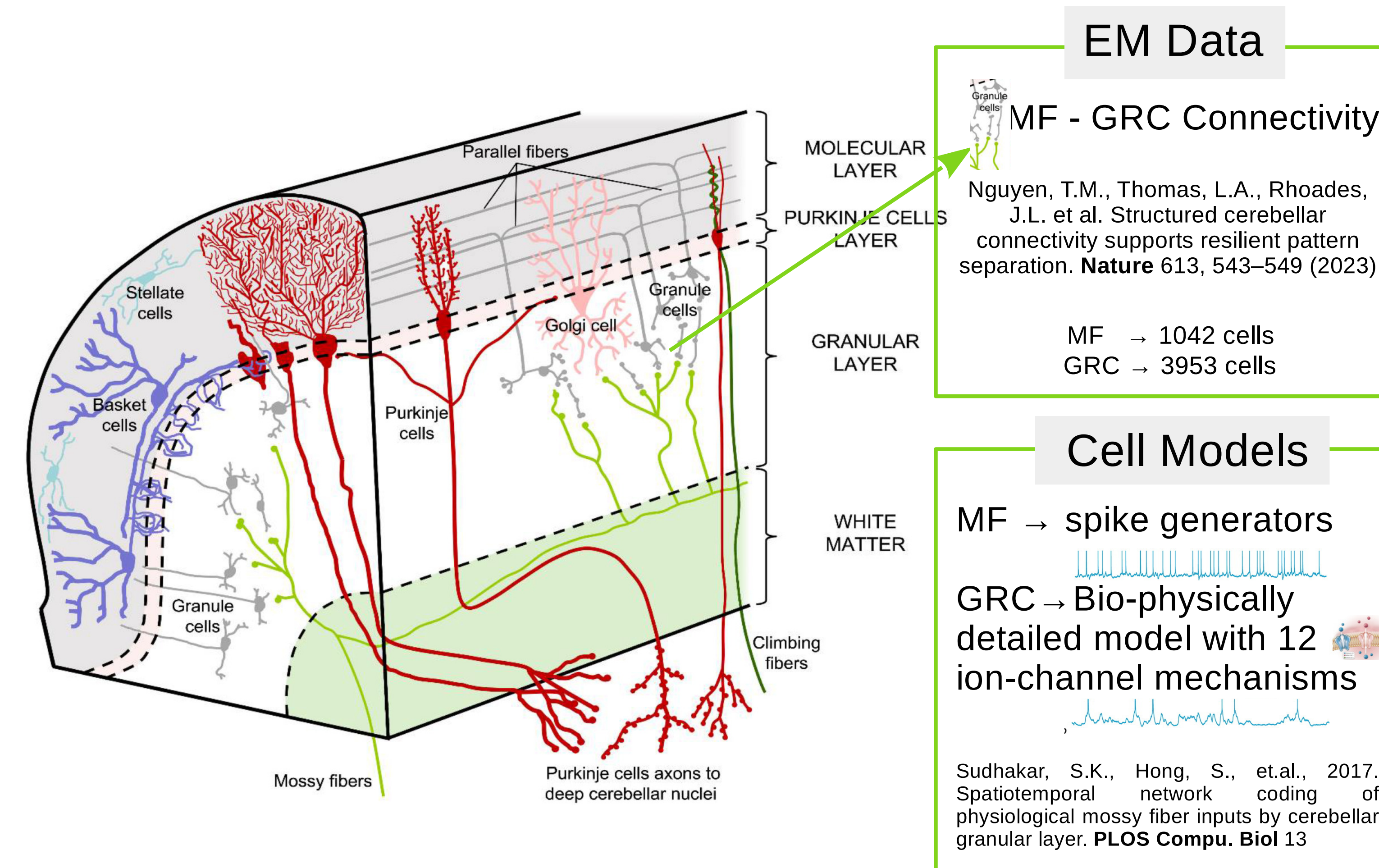
## Containerization for Scientific Computing

A **container** is a **lightweight, standalone, virtualized executable environment** that includes all dependencies required to run large-scale scientific models.

## Our Computational Framework



## Benchmarking Data and Network: Cerebellar Mossy Fiber → Granule Cell



We run multiple copies ( $n = 280$ ) of the network connectivity model simultaneously to simulate extreme computational workloads.

This allows us to rigorously assess the scalability and performance limits of GPUs under high-demand conditions.

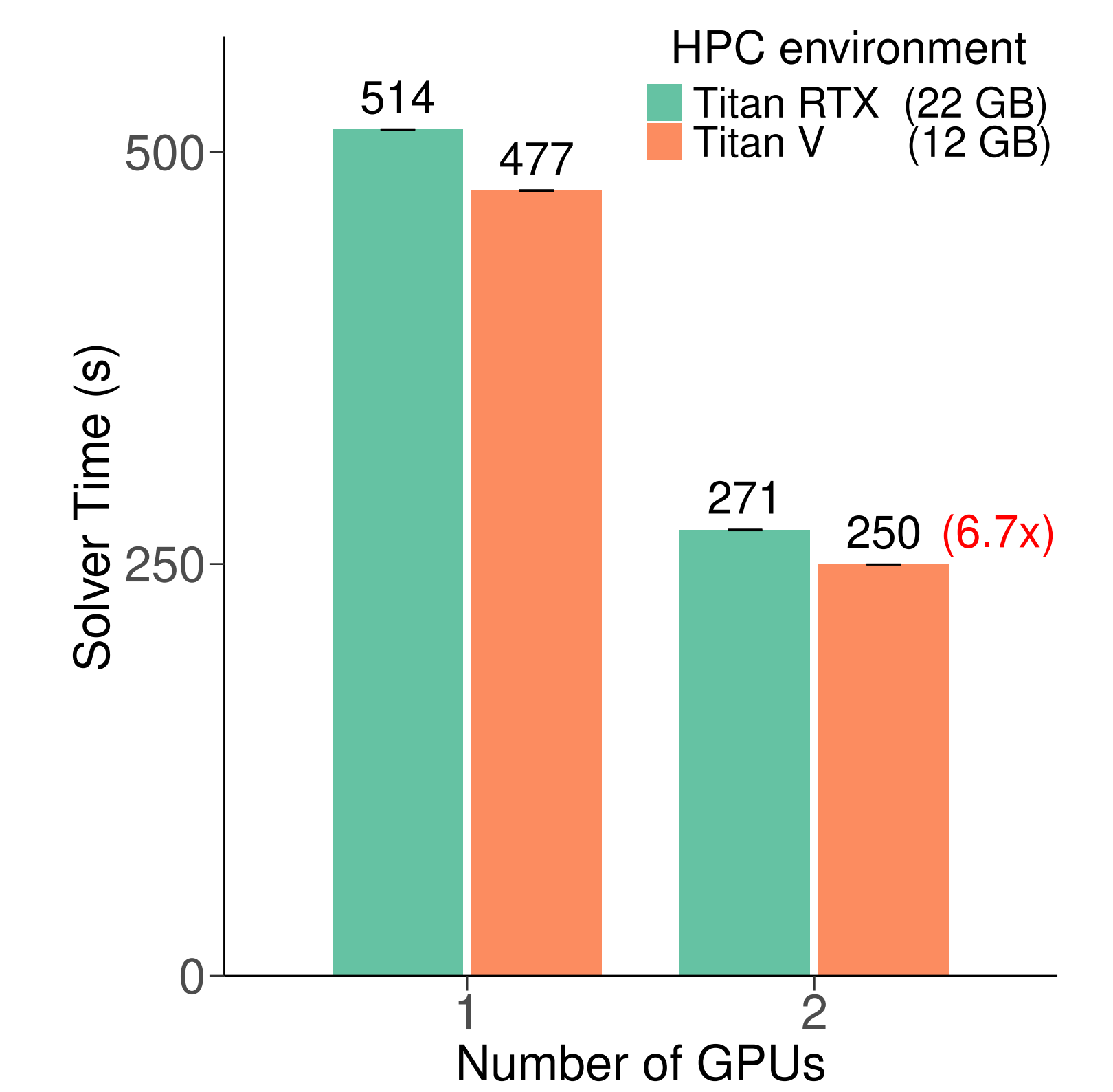
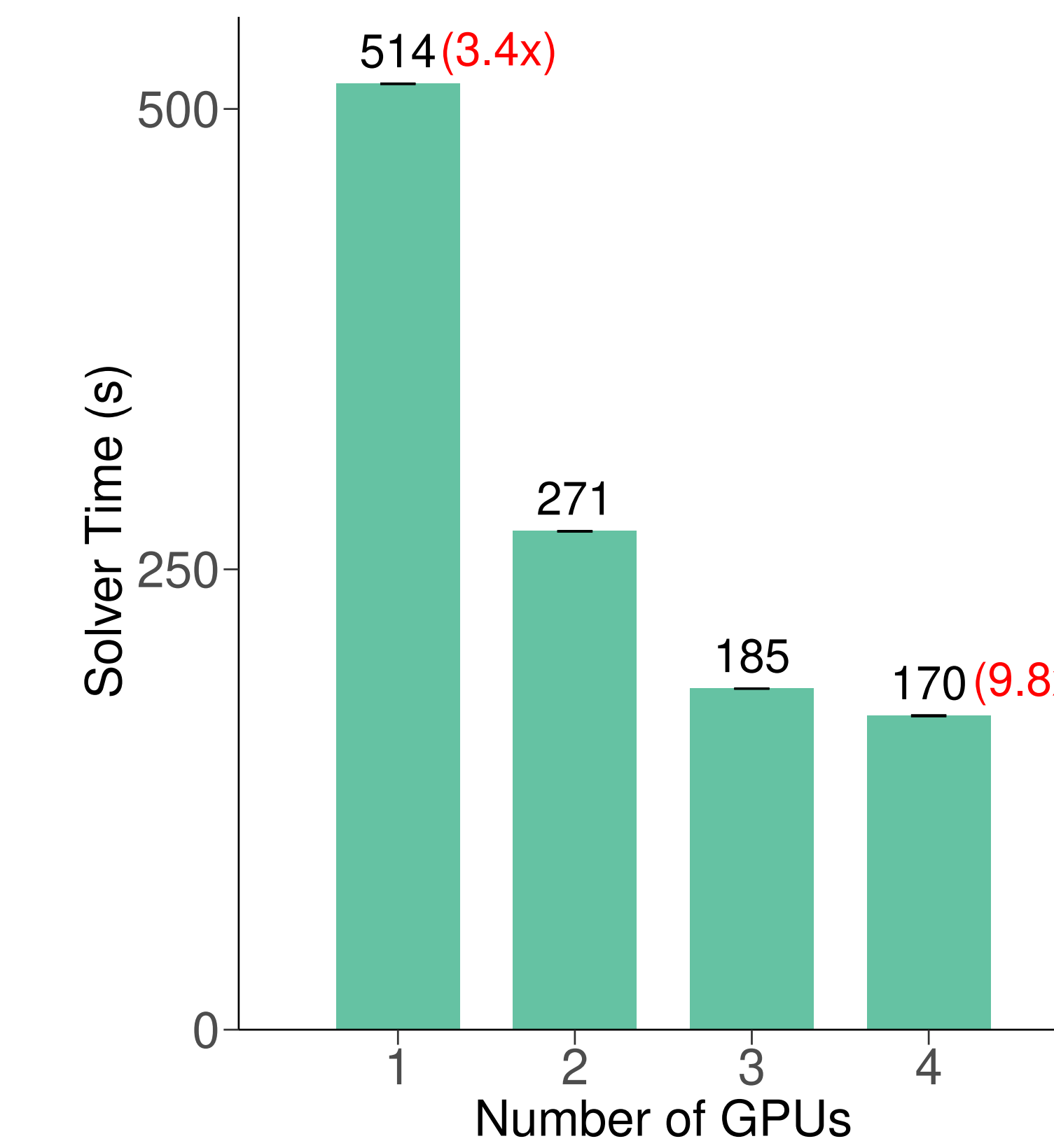
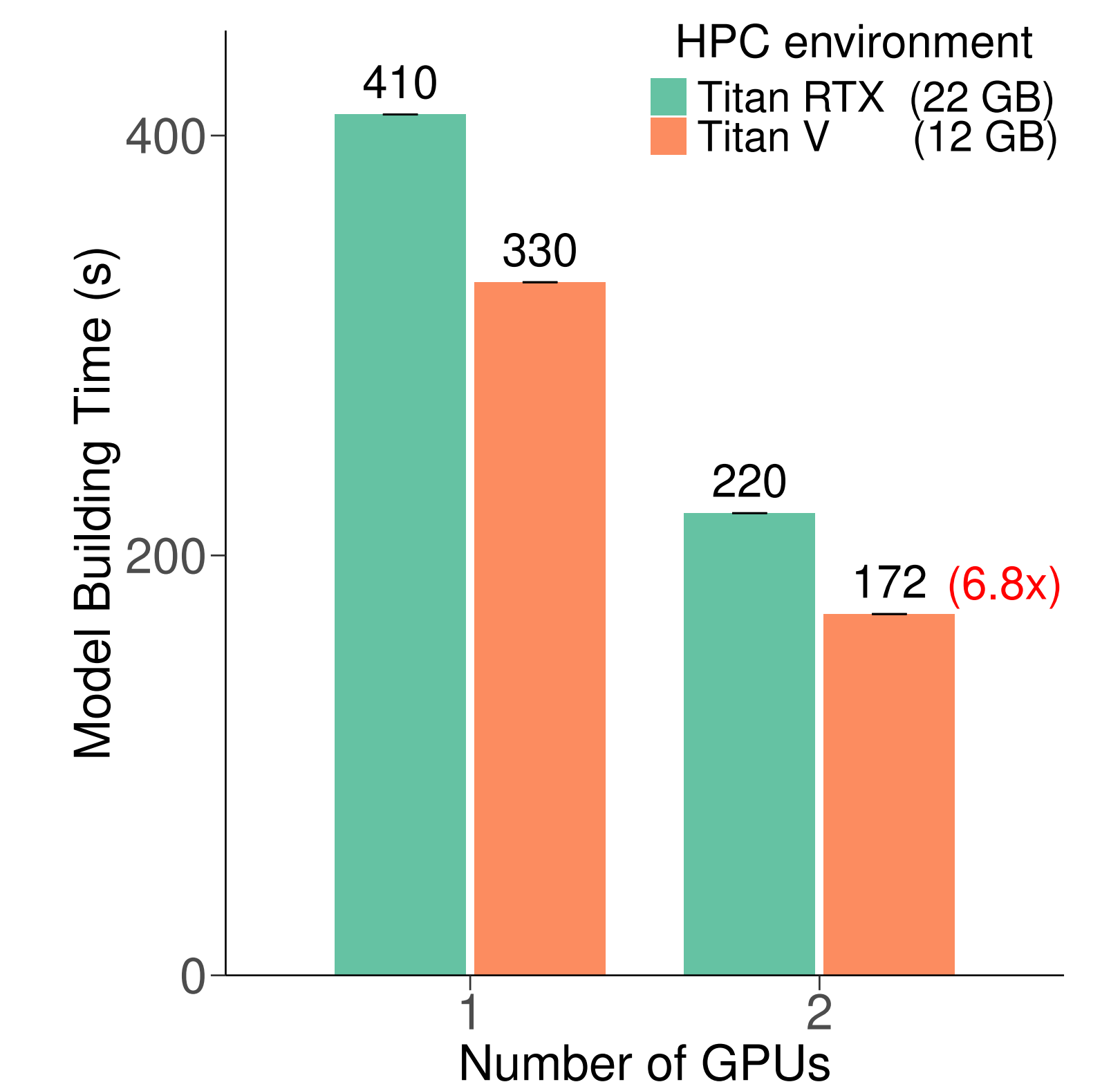
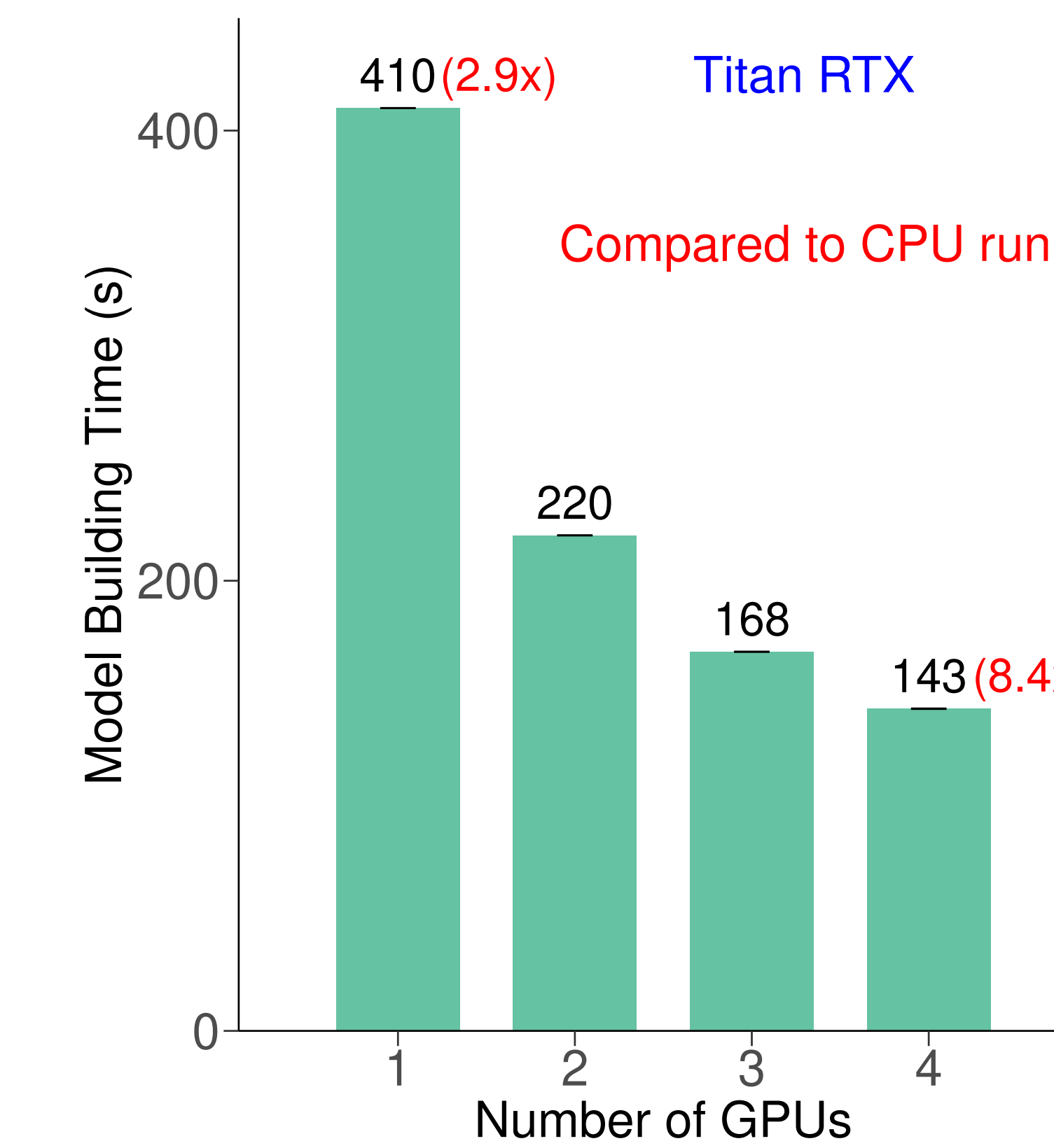
## Benchmarking HPC Environments

Table: Network Configuration

| Network elements | Count (millions) |
|------------------|------------------|
| MF               | 0.299            |
| GRC              | 1.099            |
| Compartments     | 3.297            |
| Synapses         | 3.468            |

Table: Simulation Setup

|                     |            |
|---------------------|------------|
| Simulation time     | 1 s        |
| Simulation software | CoreNEURON |



TITAN RTXs outperform 30-core CPUs by up to 8.4x (on model building) 9.8x (on solver time), with performance improving as GPU count increases.

## Future works:

🔧 **Develop a biologically realistic cerebellar circuit model** including granule layer (with Golgi cells), Purkinje layer, and other interneurons.

🔧 **Benchmark network performance across multiple institutions** to evaluate and understand the scalability and performance limits of HPC systems.

## Acknowledgment

This work was supported by IBS-R001-D2-2025-a004D5

